

Render-Timelapse

Record your Blender editing process and create a beautiful video of it.



Digital Artists

Showcase & review
creation process on
social media

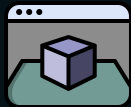


Tutors

Create step-by-step
visualization for
courses



Add-on



Rendered Output

No Blender UI. Stylized
rendering of primitives



Action Filtering

Skip experimental & faulty
steps via undotree



Smart Camera

Smooth interpolated
viewport path



Timelapse Recording

Capture editor/viewport with fixed/rotating camera after some user actions and create video from pictures.

- ScreenSnap Turntable (designersoup.itch.io)
- Timelapse Extension (extensions.blender.org)
- ReelFlow-Pro (superhivemarket.com)
- Eastape Timelapse Pro (eastapefilms.com)



Camera Positioning

Translate WASD movements into smooth bezier curves.

- AutoCam (extensions.blender.org)

Automatic extraction of view points, paths, and surfaces.

- History Assisted View Authoring (doi.org/10.1145/2556288.2557009)



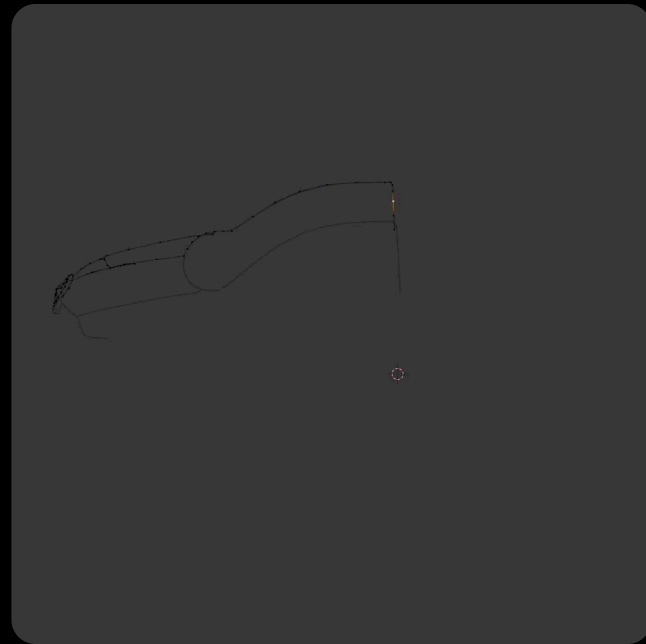
Action Recording

Record sequence of operators to review or repeat later.

- Action Recorder (github.com/InamuraJIN/ActionRecorder)
- Macros Recorder (github.com/dairin0d/macros-recorder)
- Chronicle (doi.org/10.1145/1866029.1866054)
- 3D Timeline (doi.org/10.1111/cgf.12311)

Limitations of ScreenSnap

- Captures viewport in specific shading mode, incl. "Blender Look" for primitives & mode switches
- Captures all actions, even undone actions
- Simplistic camera movement independent of actions



ScreenSnap Example Output

youtube.com/watch?v=iDGZUz5b444

Workflow



User Actions

- 🖥 Viewport
- 📦 Objects
- ✂ Edit Operations
- 🔪 Sculpting
- 🧠 Materials, Shading
- 🔗 Rigging
- 🔧 Modifiers, Geometry Nodes
- ↶ Undo/Redo
- 📄 Scripts, Add-ons
- 🎮 *Mode*
- ⊠ *Physics, Particles*
- 📁 *Workspace, UI*
- ▶ *Animation*

Techniques

- Operator-Polling, e.g. `bpy.context.window_manager.operators[i]`
- Depsgraph-Update, e.g. `bpy.app.handlers.depsgraph_update_post`
- Scene/Data Snapshot, e.g. `bpy.context.active_object`
- Viewport Draw, e.g. `bpy.types.SpaceView3D.draw_handler_add`
- Undo/Redo, e.g. `bpy.app.handlers.undo_pre`
- Rendering, e.g. `bpy.app.handlers.render_complete`
- Keylogging, e.g. `modal_handler_add`*

Phase 1: Core MVP Functionalities

Jun 17 → Jul 1 (Milestone Presentation)

^v Track actions (operators, viewport, scene)

🗄 Recording format & storage

🌿 Filter via undo-tree

📷 Basic camera interpolation

📺 Render video from recording in new blender project

⚙ UI panel

→ Working add-on with scoped core features and a demo video.

Phase 2: Polish & Stretch Goals

Jul 1 → Jul 15 (Final Presentation & Demo)

- 🕒 Time buffer for unexpected issues
- ✎ Polish UX & edge cases with sensible defaults
- 🕒 Adaptive frame timing
- 📷 Improved camera interpolation
- ⚡ Performance optimizations
- 🔗 Vertex/edge overlays

→ Polished demo video & final presentation ready.

Further Ideas:

Action Inspector

Workflow Analysis

Semantic grouping of actions & scene components

Action animations

Uncertainties

- 🗄 **Recording Format:** JSON/text initially, binary format for less storage. Representation of undo tree.
- 📷 **Camera Interpolation:** Linear, spline, or ML-based approximation of viewport path.
- 🏠 **Scene Types:** Landscape vs. orbitable. No viewport change during scripts/geometry nodes.
- ⌚ **Timing Model:** Adaptive intervals, filter irrelevant actions, pauses for camera movement, script execution.
- ⚡ **Performance Hit:** Add-on overhead running constantly in background. Render speed for large or long-edited scenes.

Q&A

Who has edited a model and wished for a timelapse tool?

What would you use it for? What's missing in existing tools?



github.com/moellh/blender-render-timelapse

